1. Import Libraries

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_scor
```

2. Load Dataset

```
url = "https://raw.githubusercontent.com/nunnarilabs/ml
df = pd.read_csv(url)

df.head()
```

	id	date	price	bedrooms	bathrooms	sqft_living	s
0	7129300520	20141013T000000	221900.0	3	1.00	1180	
1	6414100192	20141209T000000	538000.0	3	2.25	2570	
2	5631500400	20150225T000000	180000.0	2	1.00	770	
3	2487200875	20141209T000000	604000.0	4	3.00	1960	
4	1954400510	20150218T000000	510000.0	3	2.00	1680	

5 rows × 21 columns

3. Data Preparation

```
df = df.dropna()

# Features & target
y = df['price']
```

4. Train-Test Split

```
# For simple regression
X_simple = df[['sqft_living']]

X_train_s, X_test_s, y_train_s, y_test_s = train_test_split(X_simple, y, tes
```

5. Simple Linear Regression

```
model_simple = LinearRegression()
model_simple.fit(X_train_s, y_train_s)

y_pred_s = model_simple.predict(X_test_s)
```

6. Multiple Linear Regression

```
X_multi = df[['sqft_living', 'bedrooms', 'bathrooms']]

X_train_m, X_test_m, y_train_m, y_test_m = train_test_split(X_multi, y, test

model_multi = LinearRegression()
model_multi.fit(X_train_m, y_train_m)

y_pred_m = model_multi.predict(X_test_m)
```

7. Polynomial Regression

```
poly = PolynomialFeatures(degree=2)

X_poly = poly.fit_transform(X_simple)

X_train_p, X_test_p, y_train_p, y_test_p = train_test_split(X_poly, y, test_

model_poly = LinearRegression()
model_poly.fit(X_train_p, y_train_p)

y_pred_p = model_poly.predict(X_test_p)
```

8. Model Evaluation

```
print("Simple Linear Regression:")
print("MSE:", mean_squared_error(y_test_s, y_pred_s))
print("MAE:", mean_absolute_error(y_test_s, y_pred_s))
print("R2 Score:", r2_score(y_test_s, y_pred_s))

print("\nMultiple Linear Regression:")
print("MSE:", mean_squared_error(y_test_m, y_pred_m))
```

```
print("MAE:", mean_absolute_error(y_test_m, y_pred_m))
print("R2 Score:", r2_score(y_test_m, y_pred_m))

print("\nPolynomial Regression:")
print("MSE:", mean_squared_error(y_test_p, y_pred_p))
print("MAE:", mean_absolute_error(y_test_p, y_pred_p))
print("R2 Score:", r2_score(y_test_p, y_pred_p))
```

```
Simple Linear Regression:
MSE: 76484977061.77612
MAE: 177867.54034434858
R2 Score: 0.49406905389089006

Multiple Linear Regression:
MSE: 74237634953.1832
MAE: 174662.93438524107
R2 Score: 0.5089347172264876

Polynomial Regression:
MSE: 82113931183.64449
MAE: 171786.30972476854
R2 Score: 0.45683478653690435
```

9. Conclusion

- Multiple Linear Regression performs better than Simple Linear Regression
- Polynomial Regression captures non-linear patterns
- Using more relevant features improves prediction accuracy